The University of Sydney

MOOC Introduction to Calculus

Notes for 'Areas bounded by curves'

Important Ideas and Useful Facts:

(i) Areas of rectangles and triangles: The area of a rectangle of side lengths a and b is the product ab, and this can be taken as a geometric definition of multiplication of nonnegative real numbers. The area of a triangle of base length b and height h is $\frac{bh}{2}$.

(ii) Perimeter and area of a circle: Consider a circle of radius r. The perimeter P is given by the formula $P = 2\pi r$ and the area A by the formula $A = \pi r^2$. Notice that $\frac{dA}{dr} = P$.

(iii) Sigma notation: A sum of numbers, typically with many terms, is often abbreviated using *sigma notation*, employing the Greek capital $\sum$ for 'sum':

$$a_m + a_{m+1} + a_{m+2} + \ldots + a_n \;=\; \sum_{i=m}^{n} a_i \,.$$

The general term or expression is indicated by a_i, say, where the subscript i ranges from m to n, say, in steps of one at a time, where $m \leq n$. The smallest value of the subscript m appears below the $\sum$ symbol and the largest value n appears above the symbol.

For example, if $a_i = i$, $m = 1$ and $n = 100$, then

$$\sum_{i=1}^{100} i \;=\; 1 + 2 + 3 + \ldots + 100 \qquad \text{and} \qquad \sum_{i=1}^{100} i^2 \;=\; 1^2 + 2^2 + 3^2 + \ldots + 100^2 \,,$$

the sums of the first 100 consecutive integers, and squares of consecutive integers, respectively. We have the following simple but useful properties, which allow one to break a sum into pieces and to bring constants outside the sum, where $b_m, \ldots, b_n$ form any other list of numbers and c is a constant:

$$\sum_{i=m}^{n} (a_i + b_i) \;=\; \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i \qquad \text{and} \qquad \sum_{i=m}^{n} c a_i \;=\; c \sum_{i=m}^{n} a_i \,.$$

(iv) Trick of Gauss: To add together a list of evenly spaced numbers (an *arithmetic sequence*), one can add them twice, but with one list backwards, and then divide by two. For example, consider the following sum, denoted by S:

$$\begin{array}{ccccccccccccc}
1 & + & 2 & + & 3 & + & \ldots & + & n-1 & + & n & = & S \\
n & + & n-1 & + & n-2 & + & \ldots & + & 2 & + & 1 & = & S \\
\hline
n+1 & + & n+1 & + & n+1 & + & \ldots & + & n+1 & + & n+1 & = & 2S
\end{array}$$

We therefore get $2S = n(n+1)$, so that, dividing by 2, and also using $\sum$ notation,

$$\sum_{i=1}^{n} i \;=\; 1 + 2 + \ldots + n \;=\; \frac{n(n+1)}{2} \,.$$

For example,

$$\sum_{i=1}^{100} i \;=\; 1 + 2 + \ldots + 100 \;=\; \frac{100 \times 101}{2} \;=\; 5050 \,.$$

Examples:

1. Use the trick of Gauss to add up the first 50 consecutive integers:

$$1 + 2 + 3 + \ldots + 49 + 50 \; .$$

Solution: Call this sum S. Then $2S = 50 \times 51$, so $S = \dfrac{50 \times 51}{2} \; = \; 1,275$.

2. Express the following sum using sigma notation, and evaluate it:

$$1 + 2 + 3 + \cdots + 499 + 500 \; .$$

Solution: This sum becomes

$$\sum_{i=1}^{500} i \; = \; \frac{500 \times 501}{2} \; = \; 125,250 \; .$$

3. Use the trick of Gauss to add up the first 50 consecutive even integers:

$$2 + 4 + 6 + \ldots + 98 + 100 \; .$$

Solution: Call this sum S. Then $2S = 50 \times 102$, since there are 50 terms and, when adding them twice, one can group each pair to add to 102, so $S = \frac{50 \times 102}{2} \; = \; 2,550$.

Alternatively, using sigma notation, and the formula for the sum of consecutive integers,

$$S \; = \; \sum_{i=1}^{50} 2i \; = \; 2 \sum_{i=1}^{50} i \; = \; \frac{2 \times 50 \times 51}{2} \; = \; 50 \times 51 \; = \; 2,550 \; .$$

4. Evaluate $\displaystyle\sum_{i=2}^{6} 2$.

Solution: This is an abbreviation for repeated addition of 2 using 5 occurrences (for $i = 2, 3, 4, 5, 6$), so the sum become $5 \times 2 = 10$.

5. Evaluate $\displaystyle\sum_{i=2}^{6} (3i - 2)$.

Solution: This is an abbreviation for the following sum:

$$\begin{aligned} \sum_{i=2}^{6} (3i - 2) \; &= \; (3 \times 2 - 2) + (3 \times 3 - 2) + (3 \times 4 - 2) + (3 \times 5 - 2) + (3 \times 6 - 2) \\ &= \; 4 + 7 + 10 + 13 + 16 \; = \; 50 \; . \end{aligned}$$

Alternatively, exploiting the formula for the sum of the first six consecutive integers,

$$\sum_{i=2}^{6} (3i - 2) \; = \; \left(3 \sum_{i=2}^{6} i \right) - \left(\sum_{i=2}^{6} 2 \right) \; = \; 3 \times \left(\frac{6 \times 7}{2} - 1 \right) - (5 \times 2) \; = \; 60 - 10 \; = \; 50 \; .$$

6. Evaluate $\sum_{i=1}^{100}(3i-2)$.

Solution: Using properties of $\sum$ notation, we have

$$\sum_{i=1}^{100}(3i-2) \;=\; \left(3\sum_{i=1}^{100} i\right) - \left(\sum_{i=1}^{100} 2\right) \;=\; \frac{3\times 100\times 101}{2} - 200 \;=\; 14,950\ .$$

7. Find n such that $\sum_{i=1}^{n} i = 630$.

Solution: We want n such that

$$630 \;=\; \sum_{i=1}^{n} i \;=\; \frac{n(n+1)}{2}\ ,$$

so that $n^2+n = 1260$, that is,

$$n^2+n-1260 \;=\; 0\ .$$

By the quadratic formula,

$$n \;=\; \frac{-1\pm\sqrt{1+(4\times 1260)}}{2} \;=\; \frac{-1\pm\sqrt{5041}}{2} \;=\; \frac{-1\pm 71}{2}\ .$$

Taking the positive square root, we get $n = 35$.

8. Evaluate $\sum_{i=1}^{200}\left(\frac{1}{i}-\frac{1}{i+1}\right)$.

Solution: This is an example of a *collapsing sum*, since, if one writes it out in full, all of the intermediate terms cancel out leaving only the fraction $\frac{1}{1}$ at the beginning and the negative fraction $-\frac{1}{201}$ at the end:

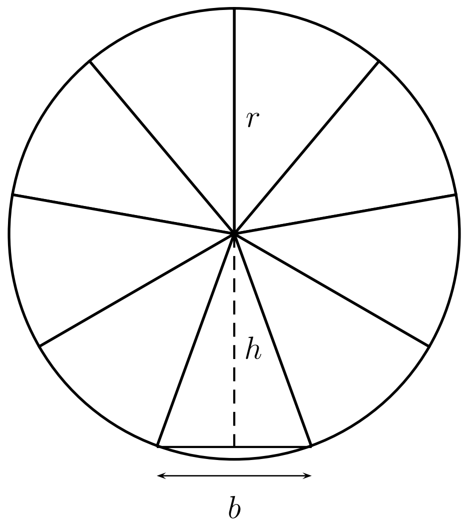
$$\sum_{i=1}^{200}\left(\frac{1}{i}-\frac{1}{i+1}\right) \;=\; \frac{1}{1}-\frac{1}{201} \;=\; \frac{200}{201}\ .$$

One can also see this more formally as follows, using properties of sigma notation, and replacing $i+1$ by j in the second sum, and noting that j goes from 2 to 201 as i goes from 1 to 200:

$$\sum_{i=1}^{200}\left(\frac{1}{i}-\frac{1}{i+1}\right) \;=\; \left(\sum_{i=1}^{200}\frac{1}{i}\right)-\left(\sum_{i=1}^{200}\frac{1}{i+1}\right) \;=\; \left(\sum_{i=1}^{200}\frac{1}{i}\right)-\left(\sum_{j=2}^{201}\frac{1}{j}\right)$$

$$=\; \frac{1}{1}+\left(\sum_{i=2}^{200}\frac{1}{i}\right)-\left(\sum_{j=2}^{200}\frac{1}{j}\right)-\frac{1}{201} \;=\; \frac{1}{1}+0-\frac{1}{201} \;=\; \frac{200}{201}\ .$$

9. We explain the Greek thought experiment that deduces the formula for the area A of a circle of radius r from the perimeter $P = 2\pi r$. We divide a circle of radius r into n sectors all of exactly the same size and shape.



The area of each sector is approximated by a triangle with base b and height h, say, and then

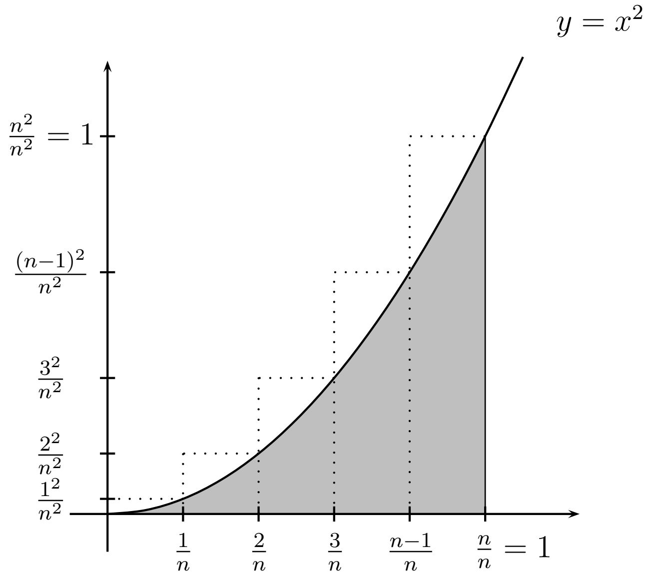
$$A \;\approx\; n \times (\text{area of one triangle}) \;=\; n \times \frac{bh}{2} \;=\; \frac{nbh}{2}\;.$$

As $n \to \infty$ the approximation improves, approaching equality, and in the limit we have,

$$A \;=\; \lim_{n\to\infty} \frac{nbh}{2} \;=\; \frac{1}{2}\big(\lim_{n\to\infty} nb\big)\big(\lim_{n\to\infty} h\big) \;=\; \frac{1}{2}\big(2\pi r\big)(r) \;=\; \pi r^2\;,$$

noting that nb approaches the perimeter $P = 2\pi r$, and h approaches the radius r.

10. We will use an approximation method involving rectangles to find the area A under the parabola $y = x^2$ for $0 \le x \le 1$, by which we mean the area shaded below, between the curve, the x-axis and the vertical line $x = 1$.



Upper rectangles that sit above the curve suffice for our argument. We divide the interval $[0, 1]$ on the x-axis into n equal subintervals, each of width $\frac{1}{n}$. We want to think of n as getting arbitrarily large (but, for the convenience of interpreting the diagram above, $n = 5$).

As i goes from 1 to n, the rectangle sitting above the curve, over the subinterval $\left[\frac{i-1}{n}, \frac{i}{n}\right]$, has height $\frac{i^2}{n^2}$ and width $\frac{1}{n}$, so has area $\left(\frac{1}{n}\right)\left(\frac{i^2}{n^2}\right) = \frac{i^2}{n^3}$.

The area A under the curve is approximated by adding together the areas of all of these rectangles, which can be expressed succinctly using sigma notation, noting the common factor of $\frac{1}{n^3}$:

$$A \approx \frac{1^2}{n^3} + \frac{2^2}{n^3} + \frac{3^2}{n^3} + \ldots + \frac{(n-1)^2}{n^3} + \frac{n^2}{n^3} = \frac{1}{n^3}\sum_{i=1}^{n} i^2 .$$

As $n \to \infty$ this approximation improves toward equality and we get the following limit:

$$A = \lim_{n\to\infty} \frac{1}{n^3}\sum_{i=1}^{n} i^2 .$$

It is far from obvious how to evaluate this limit. There is a trick involving looking at n^3 in a clever way, called a *telescoping sum*. For example, a 'telescope' for 5^3 looks like the following:

$$\begin{aligned} 5^3 &= 5^3 - 4^3 + 4^3 - 3^3 + 3^3 - 2^3 + 2^3 - 1^3 + 1^3 - 0^3 \\ &= (5^3 - 4^3) + (4^3 - 3^3) + (3^3 - 2^3) + (2^3 - 1^3) + (1^3 - 0^3) . \end{aligned}$$

In general, for any large positive integer n, we have the following 'telescope', which can be manipulated using sigma notation, and making use of the formula for the sum of consecutive integers:

$$\begin{aligned} n^3 &= n^3 - (n-1)^3 + (n-1)^3 - (n-2)^3 + \ldots - 2^3 + 2^3 - 1^3 + 1^3 - 0^3 \\ &= \sum_{i=1}^{n} \big(i^3 - (i-1)^3\big) = \sum_{i=1}^{n} \big(i^3 - (i^3 - 3i^2 + 3i - 1)\big) = \sum_{i=1}^{n} \big(3i^2 - 3i + 1\big) \\ &= 3\left(\sum_{i=1}^{n} i^2\right) - 3\left(\sum_{i=1}^{n} i\right) + \left(\sum_{i=1}^{n} 1\right) = 3\left(\sum_{i=1}^{n} i^2\right) - \frac{3n(n+1)}{2} + n . \end{aligned}$$

Rearranging this, we get the following elegant expression for the sum of squares of consecutive integers:

$$\begin{aligned} \sum_{i=1}^{n} i^2 &= \frac{1}{3}\left(n^3 + \frac{3n(n+1)}{2} - n\right) = \frac{1}{6}\big(2n^3 + 3n^2 + 3n - 2n\big) \\ &= \frac{n}{6}\big(2n^2 + 3n + 1\big) = \frac{n(n+1)(2n+1)}{6} . \end{aligned}$$

We can now return to finding the area A under the parabola, by evaluating the limit we found previously, using this formula and our usual tricks for manipulating limits:

$$\begin{aligned} A &= \lim_{n\to\infty} \frac{1}{n^3}\sum_{i=1}^{n} i^2 = \lim_{n\to\infty} \frac{1}{n^3}\left(\frac{n(n+1)(2n+1)}{6}\right) = \lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} \\ &= \lim_{n\to\infty} \frac{(1+\frac{1}{n})(2+\frac{1}{n})}{6} = \frac{(1+0)(2+0)}{6} = \frac{1}{3} . \end{aligned}$$